LEDGERLINE

Documentation Pack

5 documents

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CONFIDENTIAL

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API Reference

Overview

Base URL	`http://localhost:8000` (`Dockerfile`, `docker-compose.yml`)
Format	JSON
Source	Route detection in `app/main.py` and `app/routers/orders.py` (medium confidence), confirmed by reading the code
Version	0.9.2 (`FastAPI(version=...)` in `app/main.py`)

FastAPI also serves the generated OpenAPI schema at `/openapi.json` and interactive docs at `/docs` UNTESTED.

Authentication

Every `/orders` endpoint requires an API key in the `x-api-key` header (`dependencies=[Depends(require_api_key)]` in `app/routers/orders.py`). The key is compared with the `API_KEY` environment variable (`app/auth.py`).

```
x-api-key: <your API key>
```

Situation	Response
Header missing	`422` validation error (header is required)
Wrong key	`401` `{"detail": "invalid API key"}`
`API_KEY` not set on the server	`500`: `os.environ["API_KEY"]` raises `KeyError`

Endpoints

Method	Path	Summary	Auth
`GET`	`/health`	Liveness check	no
`GET`	`/orders`	List orders	yes
`POST`	`/orders`	Create an order	yes
`GET`	`/orders/{order_id}`	Get one order	yes
`DELETE`	`/orders/{order_id}`	Cancel an order	yes

`GET /health`

Returns service status (`app/main.py`).

```
curl http://localhost:8000/health
```

Response `200`

```
{"status": "ok"}
```

`GET /orders`

Lists orders (`list_orders`, `app/routers/orders.py`).

Name	In	Type	Required	Description
`limit`	query	int	no	Maximum results, default `50`

```
curl "http://localhost:8000/orders?limit=10" -H "x-api-key: $API_KEY"
```

Response `200`: an array of `OrderOut`. The current implementation always returns `[]`.

`POST /orders`

Creates an order from an `OrderIn` body (`app/schemas.py`).

```
curl -X POST http://localhost:8000/orders \
  -H "x-api-key: $API_KEY" -H "Content-Type: application/json" \
  -d '{"customer_id": 42, "sku": "MUG-001", "quantity": 12}'
```

> **WARNING**
>
> Not implemented yet: the handler raises `501 {"detail": "not implemented"}`. The declared success status is `201`.

`GET /orders/{order_id}`

Name	In	Type	Required	Description
`order_id`	path	int	yes	Order ID

The current handler always returns `404 {"detail": "order not found"}`.

`DELETE /orders/{order_id}`

Cancels an order. Returns `204 No Content`.

```
curl -X DELETE http://localhost:8000/orders/42 -H "x-api-key: $API_KEY"
```

Schemas

OrderIn

Field	Type	Required
`customer_id`	int	yes
`sku`	string	yes
`quantity`	int	yes

OrderOut: every `OrderIn` field, plus:

Field	Type	Notes
`id`	int	Order ID
`status`	string	`pending` by default in the database (`app/models.py`)

Error codes

Status	Detail	When
401	`invalid API key`	`x-api-key` doesn't match `API_KEY`
404	`order not found`	`GET /orders/{order_id}`
422	validation error	missing header, or a body or path value of the wrong type
501	`not implemented`	`POST /orders`

Related documents

Data Model · Runbook · Integration Guide

Assumptions and open questions

#	Item	Type	Why unverified	Who can confirm
A1	`/docs` and `/openapi.json` are reachable	assumption	FastAPI default, not run	backend lead
Q1	When will `POST /orders` and `GET /orders/{order_id}` be implemented?	open question	handlers are stubs	backend lead
Q2	Should a missing `API_KEY` fail at startup instead of per request?	open question	design choice	backend lead

Budget and Resource Plan

Planning estimates only, not financial advice.

> **NOTE**
> These figures use **sample inputs** supplied to demonstrate the skill. They are not real Ledgerline data.

The plan covers running and hardening the Ledgerline API in 2027. Tooling matches what the repo uses: Sentry (`sentry-sdk`, `app/main.py`) and GitLab CI (`.gitlab-ci.yml`).

Inputs

Currency: **USD**. Period: **12 months from 2027-01**, contingency **20%**.

Input	Value	Source	Assumption
`currency`	USD	user	—
`budget_start`	2027-01	user	—
`contingency_pct`	20	user	—
`team_roles`	3 item(s)	user	—
`tooling_costs`	2 item(s)	user	—
`fixed_infra_cost`	450	user	—
`one_time_costs`	2 item(s)	user	—

Team

Role	Allocation	Monthly rate	Monthly cost	Annual cost
Backend developer	100%	USD 11,000	USD 11,000	USD 132,000
DevOps engineer	30%	USD 12,000	USD 3,600	USD 43,200
QA engineer	25%	USD 8,000	USD 2,000	USD 24,000

Infrastructure and tooling

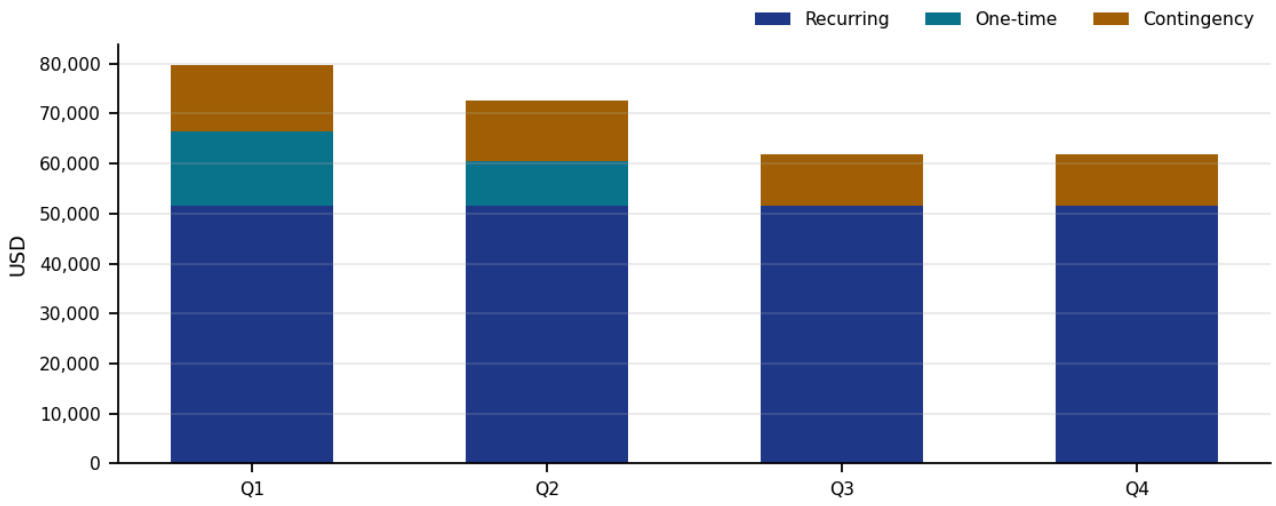
Item	Monthly	Annual
Sentry Team	USD 26	USD 312
GitLab Premium (3 seats)	USD 87	USD 1,044

Item	Monthly	Annual
Infrastructure (fixed)	USD 450	USD 5,400

Licences and one-time costs

Item	Amount	Quarter
Penetration test	USD 9,000	Q2
SOC 2 readiness assessment	USD 15,000	Q1 ASSUMPTION

Totals by quarter



Budget by quarter (USD)

Quarter	Recurring	One-time	Contingency	Total
Q1 (Jan 2027 – Mar 2027)	USD 51,489	USD 15,000	USD 13,298	USD 79,787
Q2 (Apr 2027 – Jun 2027)	USD 51,489	USD 9,000	USD 12,098	USD 72,587
Q3 (Jul 2027 – Sep 2027)	USD 51,489	USD 0	USD 10,298	USD 61,787
Q4 (Oct 2027 – Dec 2027)	USD 51,489	USD 0	USD 10,298	USD 61,787
Year	**USD 205,956**	**USD 24,000**	**USD 45,991**	**USD 275,947**

Formulas

```
team_monthly[r]   = allocation[r] × monthly_rate[r]
recurring_monthly = Σ team_monthly + Σ tooling + fixed_infra
recurring[q]      = 3 × recurring_monthly
contingency[q]    = (recurring[q] + one_time[q]) × contingency_pct
total[q]          = recurring[q] + one_time[q] + contingency[q]
```

One-time costs without a month are placed in Q1 ASSUMPTION.

Related documents

1-Year Forecast · Runbook

Assumptions and open questions

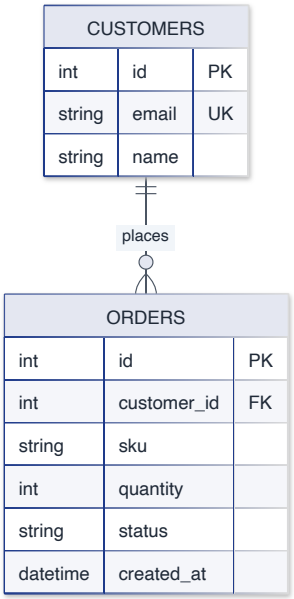
#	Item	Type	Why unverified	Who can confirm
A1	All rates and costs are sample values	assumption	demonstration run	finance
A 2	SOC 2 readiness lands in Q1	assumption	no month given	compliance owner
Q1	Who runs `scripts/deploy.sh`? It is referenced in CI but missing	open question	file not in repo	DevOps

Data Model

Overview

Database	PostgreSQL 16 (`postgres:16` in `docker-compose.yml`)
Access	SQLAlchemy 2.0 ORM, typed `Mapped[...]` models
Source	`app/models.py` (models), `alembic/versions/` (migrations)
Entities	2: `Customer`, `Order`

Entity relationship diagram



Entities

Customer

A wholesale buyer (`app/models.py`, table `customers`).

Field	Type	Null	Key	Notes
`id`	int	no	PK	
`email`	varchar(255)	no	unique	
`name`	text	no		

Order

An order line for one SKU (`app/models.py`, table `orders`).

Field	Type	Null	Key	Notes
`id`	int	no	PK	
`customer_id`	int	no	FK → `customers.id`	indexed
`sku`	varchar(64)	no		
`quantity`	int	no		
`status`	text	no		default `pending`
`created_at`	timestamp	no		default `datetime.utcnow`

Relationships

From	To	Cardinality	On delete	Evidence
Customer	Order	1:N	database default ASSUMPTION	`relationship(back_populates=...)` in `app/models.py`

Indexes and constraints

Table	Index / constraint	Columns	Type
`customers`	primary key	`id`	btree
`customers`	unique	`email`	btree
`orders`	primary key	`id`	btree
`orders`	index	`customer_id`	btree
`orders`	foreign key	`customer_id` → `customers.id`	constraint

Migrations

- Tool: Alembic · Location: `alembic/versions/` · Count: 2
- `0001_init.py`: create customers and orders
- `0002_order_status.py`: add order status

```
make migrate     # alembic upgrade head
```

> **NOTE**
>
> `created_at` uses `datetime.utcnow`, which returns naive timestamps. Store and compare times as UTC throughout.

Related documents

API Reference · Architecture

Assumptions and open questions

#	Item	Type	Why unverified	Who can confirm
A1	On-delete behaviour for orders	assumption	no `ondelete=` set in the model	backend lead
Q1	Should `status` be an enum?	open question	free text today	backend lead

1-Year Forecast

Planning estimates only, not financial advice.

> **NOTE**
> These figures use **sample inputs** supplied to demonstrate the skill. They are not real Ledgerline data.

Users here are **wholesaler accounts** calling the API. Infrastructure is the API container plus PostgreSQL 16 (`docker-compose.yml`) and S3 storage for invoices (`AWS_S3_BUCKET` in `.env.example`). The hosting provider is not in the repo.

Assumptions

Currency: **USD**. Period: **monthly, Jan 2027 – Dec 2027**.

Input	Value	Source	Assumption
`currency`	USD	user	—
`forecast_start`	2027-01	user	—
`starting_users`	180	user	—
`monthly_growth_pct`	2-6	user	—
`infra_cost_per_user`	0.6	user	—
`fixed_infra_cost`	450	user	—
`revenue_model`	subscription	user	—
`arpu`	149	user	—
`paid_conversion_pct`	20-35	user	—

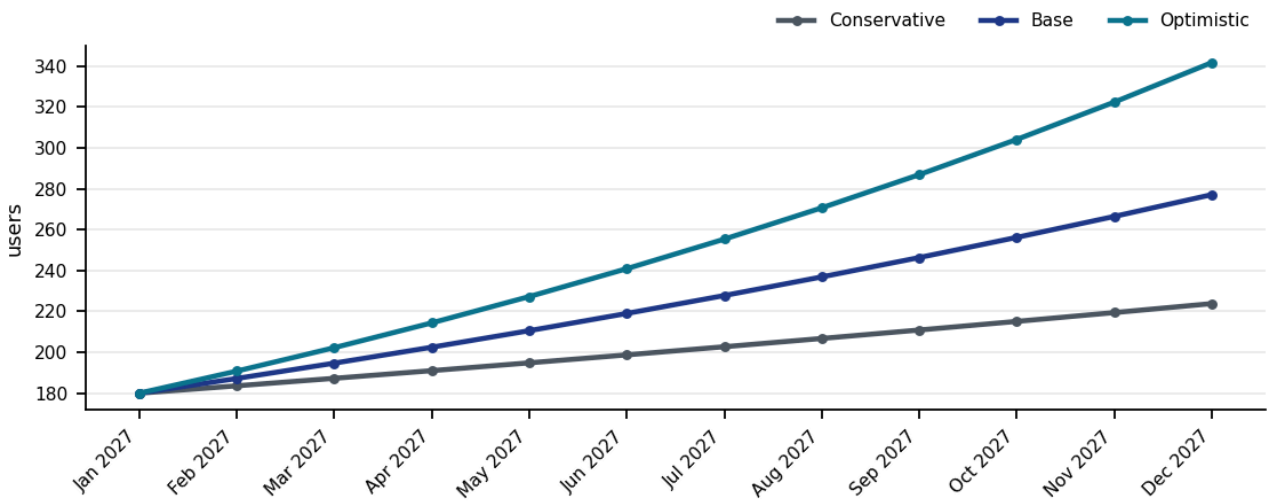
Method

```
users[m]   = starting_users × (1 + growth)^m          (m = 0..11)
infra[m]   = fixed_infra_cost + users[m] × infra_cost_per_user
paying[m]  = users[m] × paid_conversion
revenue[m] = paying[m] × arpu
net        = Σ revenue - Σ infra
```

Scenarios

Scenario	Growth / month	Conversion
Conservative	2%	20%
Base	4%	27.5%
Optimistic	6%	35%

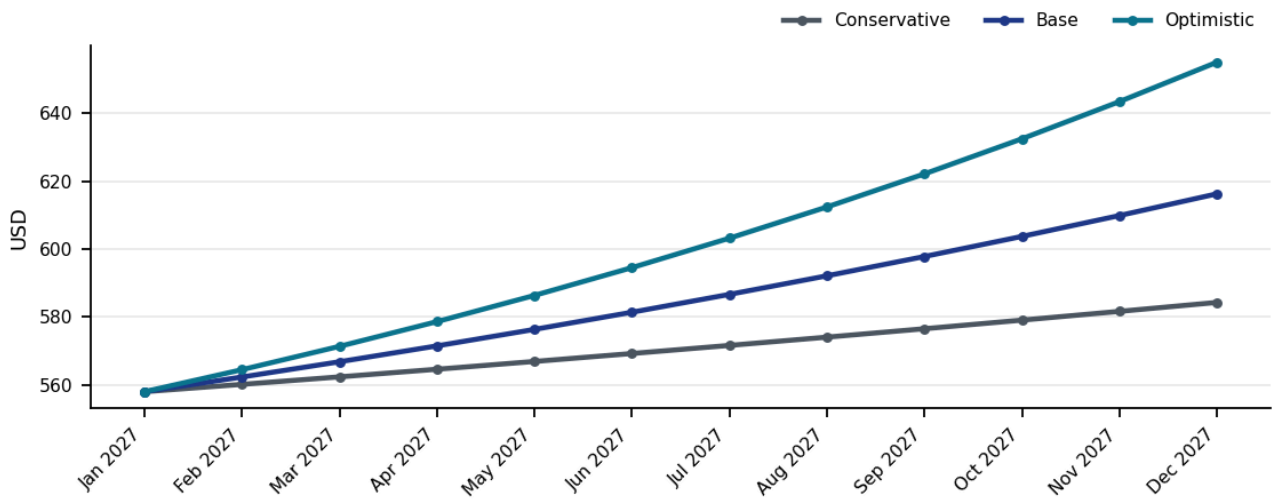
Users and usage



Active users by scenario

Month	Conservative	Base	Optimistic
Jan 2027	180	180	180
Feb 2027	184	187	191
Mar 2027	187	195	202
Apr 2027	191	202	214
May 2027	195	211	227
Jun 2027	199	219	241
Jul 2027	203	228	255
Aug 2027	207	237	271
Sep 2027	211	246	287
Oct 2027	215	256	304
Nov 2027	219	266	322
Dec 2027	224	277	342

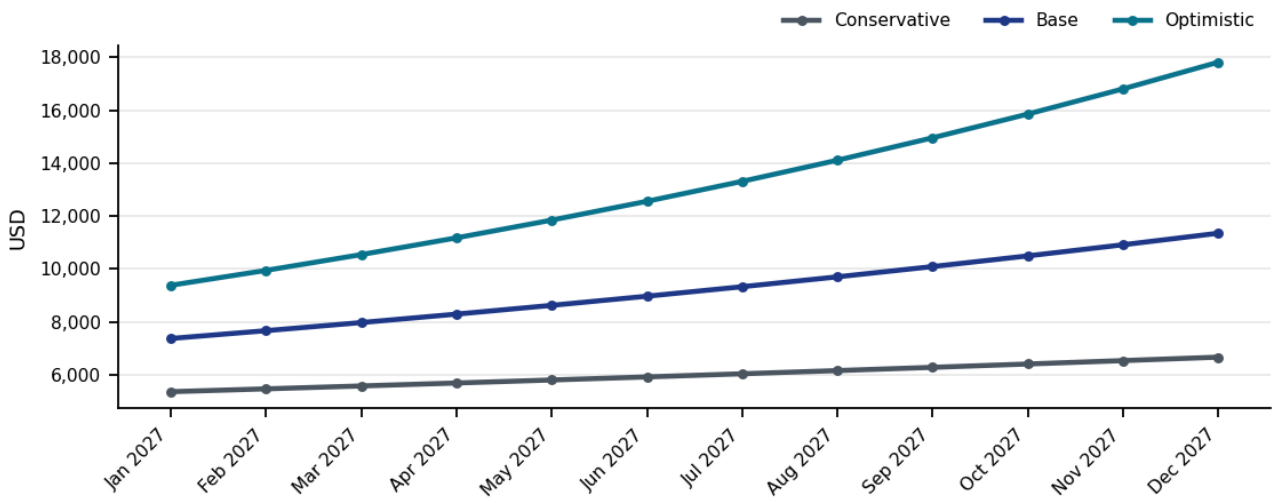
Infrastructure cost



Monthly infrastructure cost (USD)

Month	Conservative	Base	Optimistic
Jan 2027	USD 558	USD 558	USD 558
Feb 2027	USD 560	USD 562	USD 564
Mar 2027	USD 562	USD 567	USD 571
Apr 2027	USD 565	USD 571	USD 579
May 2027	USD 567	USD 576	USD 586
Jun 2027	USD 569	USD 581	USD 595
Jul 2027	USD 572	USD 587	USD 603
Aug 2027	USD 574	USD 592	USD 612
Sep 2027	USD 577	USD 598	USD 622
Oct 2027	USD 579	USD 604	USD 632
Nov 2027	USD 582	USD 610	USD 643
Dec 2027	USD 584	USD 616	USD 655

Revenue



Monthly revenue (USD)

Month	Conservative	Base	Optimistic
Jan 2027	USD 5,364	USD 7,376	USD 9,387
Feb 2027	USD 5,471	USD 7,671	USD 9,950
Mar 2027	USD 5,581	USD 7,977	USD 10,547
Apr 2027	USD 5,692	USD 8,296	USD 11,180
May 2027	USD 5,806	USD 8,628	USD 11,851
Jun 2027	USD 5,922	USD 8,973	USD 12,562
Jul 2027	USD 6,041	USD 9,332	USD 13,316
Aug 2027	USD 6,162	USD 9,706	USD 14,115
Sep 2027	USD 6,285	USD 10,094	USD 14,961
Oct 2027	USD 6,410	USD 10,498	USD 15,859
Nov 2027	USD 6,539	USD 10,918	USD 16,811
Dec 2027	USD 6,669	USD 11,354	USD 17,819

Summary

Metric (year 1)	Conservative	Base	Optimistic	Range
Active users at month 12	224	277	342	224 – 342
Total infra cost	USD 6,849	USD 7,023	USD 7,222	USD 6,849 – USD 7,222
Total revenue	USD 71,942	USD 110,823	USD 158,358	USD 71,942 – USD 158,358
Net (revenue – infra)	USD 65,094	USD 103,800	USD 151,136	USD 65,094 – USD 151,136

What would change this forecast

- Each extra percentage point of monthly growth adds about **31** users by month 12 (base scenario).
- Infrastructure cost scales linearly with users; a cheaper tier or caching lowers `infra_cost_per_user`.

Related documents

Budget and Resource Plan · API Reference

Assumptions and open questions

#	Item	Type	Why unverified	Who can confirm
A1	All forecast inputs are sample values	assumption	demonstration run	product owner
A2	Hosting provider and its pricing	assumption	not in repo (`hosting: unknown`)	DevOps
Q1	Is conversion measured per account or per seat?	open question	no billing code in repo	finance

Runbook

Service overview

Hosting	Not recorded in the repo ASSUMPTION
Components	`api` (FastAPI on uvicorn, port 8000) and `db` (PostgreSQL 16), from `docker-compose.yml`
Health endpoint	`GET /health` (`app/main.py`)
On-call	Not provided ASSUMPTION
Error tracking	Sentry, if `SENTRY_DSN` is set (`app/main.py`)

Start, stop, restart

Local or single-host deployment with Docker Compose:

```
docker compose up -d            # start api + db
docker compose stop             # stop
docker compose restart api      # restart the API only
```

Without Docker, for development:

```
make dev                        # uvicorn app.main:app --reload
```

Health checks

```
curl -fsS http://localhost:8000/health
```

Expected: `{"status":"ok"}`. This only proves the process is up. It doesn't touch the database.

Routine tasks

Task	Frequency	Command
Apply database migrations	each release	`make migrate`
Run tests with coverage	each change	`make test`
Lint	each change	`make lint`
Back up PostgreSQL	not configured	see Backup and Disaster Recovery

Logs

uvicorn writes access and error logs to stdout:

```
docker compose logs -f api
docker compose logs -f db
```

Common problems

Every `/orders` request returns 500

1. Check: `docker compose exec api printenv API_KEY`
2. Fix: set `API_KEY` in `.env` and run `docker compose up -d api`. `app/auth.py` reads `os.environ["API_KEY"]` on each request and raises `KeyError` when it's missing.

`KeyError: 'DATABASE_URL'`

1. Check: `.env` contains `DATABASE_URL` (see `.env.example`).
2. Fix: add it and restart. `app/db.py` reads it with `os.environ[...]` when the module loads.

API can't reach the database

1. Check: `docker compose ps db` shows the database as running.
2. Fix: inside Compose, the host must be `db`, not `localhost`: `postgresql+psycopg://ledger:<password>@db:5432/ledger`.

Deploy job fails in GitLab CI

The `deploy` job in `.gitlab-ci.yml` runs `./scripts/deploy.sh`, but that file isn't in the repository.

> **WARNING**
> Deploys from `main` fail until `scripts/deploy.sh` is added, or the job is changed to the real deploy command.

Scaling

`docker-compose.yml` defines one `api` container. To handle more load, run several uvicorn workers (`--workers N`) or more replicas behind a load balancer UNTESTED. The database is one PostgreSQL instance.

Escalation

No on-call contact was provided. See Incident Response once it has been written.

Related documents

API Reference · Data Model · Incident Response

Assumptions and open questions

#	Item	Type	Why unverified	Who can confirm
A1	Hosting environment	assumption	`hosting: unknown`	DevOps
A2	On-call contact	assumption	`oncall_contact: unknown`	engineering manager
A3	Scaling commands	assumption	not exercised	DevOps
Q1	What should `scripts/deploy.sh` do?	open question	referenced in CI but missing	DevOps
Q2	Should `/health` check the database?	open question	currently static	backend lead